

1. Define Data mining?

The process of Extracting useful patterns and knowledge

Tasks:

- * classification

- * clustering.

2. What is Data cleaning?

Data cleaning is the process of detecting and correcting errors or inconsistent

Two quality issue:

- * Missing values

- * Noisy data.

3. Define data integration.

Combining data from multiple sources into single dataset

Challenges:

Data Redundancy

Data inconsistency.

4. what is co-variance analysis?

measures the relationship between two variables and whether they change together.

5. if 2 stocks A and B have values, will their price rise together?

The prices rise together because both stocks generally increase simultaneously, indicating positive co-variance.

6. Explain the architecture of data mining system.

Database / Data Warehouse

Data mining engine

pattern evaluation

user interface.



7. find co-variance for $x = 4, 5, 6, 7, 8, 9$ $y = 8, 3, 7, 6, 5$

sample co-variance = -0.45 (approx) $\text{COV}(a, b) = E(A \cdot B) - \bar{A} \bar{B}$

8. Describe steps in data preprocessing.

Data cleaning

Data integration

Data reduction

Data discretization

7. what is equal frequency.

Equal frequency binning divides sorted data into bins containing the same number of data values.

10. Explain data discretization.

data discretization is process of converting continuous data into discrete intervals

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